



EEE240 FUNDAMENTALS OF DIGITAL LOGIC DESIGN

Chapter no. 2

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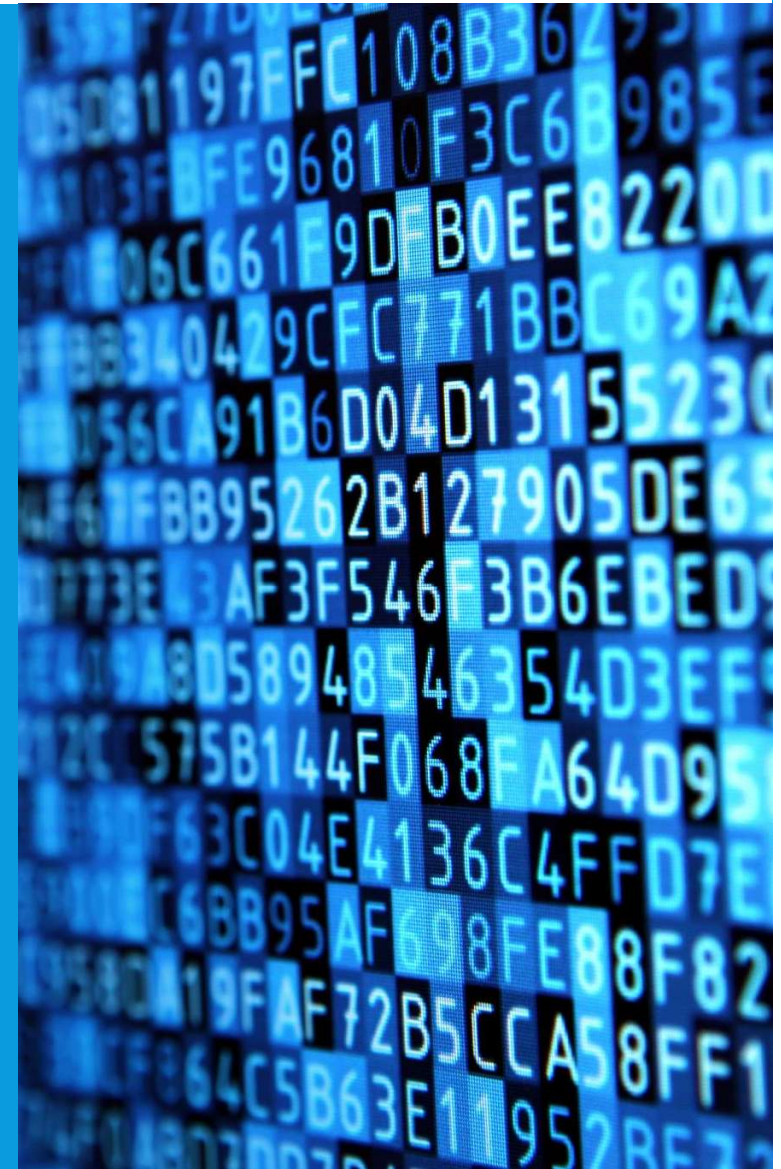
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OUTLINE

- Introduction to Boolean algebra
- Basic definition and axiomatic definition of Boolean algebra
- Boolean functions
- Canonical form
- Standard form
- Logic operations
- Digital logic gates
- Integrated circuits



MUST READ!!!

- Chapter No. 2: Boolean Algebra and Logic Gates

BASIC DEFINITIONS

$$x \in S$$

$$y \notin S$$

$$A = \{1, 2, 3, 4\}$$

- Binary Operator:
 - “*” is a binary operator iff
 - “*” is not a binary operator if

$$a * b = c;$$

$$a, b, c \in S$$

$$a * b = c;$$

$$a, b \in S, c \notin S$$

BASIC DEFINITIONS

1. Closure:

1. A set “S” is closed w.r.t. an operator iff

$$a * b = c;$$
$$a, b, c \in S$$

2. Associative Law:

$$(x * y) * z = x * (y * z); \forall x, y, z \in S$$

3. Commutative Law:

$$x * y = y * x; \forall x, y \in S$$

4. Identity Element:

$$e * x = x * e = x; \forall x \in S$$

5. Inverse:

$$x * y = e$$

6. Distributive

$$x * (y.z) = (x * y).(x * z)$$

BASIC DEFINITIONS

- Algebraic structure:
 - e.g.: Field

A field is a set of elements, together with two binary operators, each having properties 1 through 5 and both operators combining to give property 6.

- Example: The set of real numbers, together with the binary operators “+” and “.”, forms the field of real numbers. The field of real numbers is the basis for arithmetic and ordinary algebra.

BASIC DEFINITIONS

- The operators and postulates have the following meanings:
 - The binary operator “+” defines addition.
 - The additive identity is 0.
 - The additive inverse defines subtraction.
 - The binary operator “.” defines multiplication.
 - The multiplicative identity is 1.
 - For $a \neq 0$, the multiplicative inverse of $a = 1/a$ defines division (i.e., $a \cdot 1/a = 1$).
 - The only distributive law applicable is that of “.” over “+”: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

AXIOMATIC DEFINITION OF BOOLEAN ALGEBRA

- Developed by George Bool
- A useful mathematical system for specifying and transforming logic functions.
- We study Boolean algebra as a foundation for designing and analyzing digital systems

Boolean algebra is an algebraic structure defined by a set of elements, B , *together* with two binary operators, “+” and “ $\cdot$ ”, provided that the following (Huntington) postulates are satisfied:

AXIOMATIC DEFINITION OF BOOLEAN ALGEBRA

1. Closure:

- (a) The structure is closed with respect to the operator $+$.
- (b) The structure is closed with respect to the operator $\cdot$.

2. Identity Element:

- (a) The element 0 is an identity element with respect to $+$; that is, $x + 0 = 0 + x = x$.
- (b) The element 1 is an identity element with respect to $\cdot$; that is, $x \cdot 1 = 1 \cdot x = x$.

3. Commutative:

- (a) The structure is commutative with respect to $+$; that is, $x + y = y + x$.
- (b) The structure is commutative with respect to $\cdot$; that is, $x \cdot y = y \cdot x$.

4. Distributive:

- (a) The operator $\cdot$ is distributive over $+$; that is, $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$.
- (b) The operator $+$ is distributive over $\cdot$; that is, $x + (y \cdot z) = (x + y) \cdot (x + z)$.

AXIOMATIC DEFINITION OF BOOLEAN ALGEBRA

5. For every element $x \in B$, there exists an element $x' \in B$ (called the complement of x) such that
- (a) $x + x' = 1$ and
 - (b) $x \cdot x' = 0$.
6. There exist at least two elements $x, y \in B$, such that $x \neq y$.

COMPARISON OF BOOLEAN ALGEBRA WITH ARITHMETIC AND ORDINARY ALGEBRA

1. Associative Law:

- Not included in postulates
- But can be derived for both operators

2. Distributive Law:

- “+” over “•” (i.e., $x + (y \cdot z) = (x + y) \cdot (x + z)$) is valid for Boolean algebra, but not for ordinary algebra.

3. Inverse:

- No additive or multiplicative inverse
- So, no subtraction and no division operations

4. Complement:

- Ordinary algebra does not have complement

5. Undefined set of elements for Boolean algebra:

- However, two valued Boolean algebra, that we are going to use has two elements, 0 and 1

TWO-VALUED BOOLEAN ALGEBRA

- Operator tables:

x	y	$z = x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$z = x + y$
0	0	0
0	1	1
1	0	1
1	1	1

x	$z = x'$
0	1
1	0

TWO-VALUED BOOLEAN ALGEBRA

1. Is the structure closed w.r.t.
 - a) “+” (P-1(a))
 - b) “•” (P-1(b))
2. What are the identity elements?
 - a) “+” (P-2(a))
 - b) “•” (P-2(b))
3. Does it hold commutative law?
 - a) “+” (P-3(a))
 - b) “•” (P-3(b))
4. Verify the distributive law (• over +):
 - a) $x \bullet (y + z) = (x \bullet y) + (x \bullet z)$ (P-4(a))
 - b) (+ over •) (P-4(b))
5. Complement:
 - a) $x + x' = 1$ (P-5(a))
 - b) $x \bullet x' = 0$ (P-5(b))
6. Postulate 6 is also satisfied as there are two elements “0” and “1” and $0 \neq 1$. (P-6)

BOOLEAN ALGEBRA THEOREMS

- Duality

- The *dual* of a Boolean algebraic expression is obtained by interchanging the **AND** and the **OR** operators and replacing the **1**'s by **0**'s and the **0**'s by **1**'s.
- If, $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$, then using
- duality $x + (y \cdot z) = (x + y) \cdot (x + z)$

Duality is important because it doubles the usefulness of everything about Boolean algebra and manipulation of logic functions.

BOOLEAN ALGEBRA THEOREMS

▪ Theorem-1 (T-1)

$$\begin{array}{ll} \cdot & x \cdot x = x \\ & x + x = x \end{array}$$

▪ Theorem-2 (T-2)

$$\begin{array}{ll} \cdot & x \cdot 0 = 0 \\ & x + 1 = 1 \end{array}$$

▪ Theorem 3: *Involution* (T-3)

$$\cdot \quad (x')' = x$$

▪ Theorem 4: *Associative & Distributive* (T-4)

$$\begin{array}{l} \cdot \quad (x \cdot y) \cdot z = x \cdot (y \cdot z); \quad (x + y) + z = x + (y + z) \\ \cdot \quad x \cdot (y + z) = (x \cdot y) + (x \cdot z); \end{array}$$

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

▪ Theorem 5: *DeMorgan* (T-5)

$$\begin{array}{ll} \cdot & (x \cdot y)' = x' + y' \\ & (x + y)' = x' \cdot y' \end{array}$$

▪ Theorem 6: *Absorption* (T-6)

$$\begin{array}{ll} \cdot & x \cdot (x + y) = x \\ & x + (x \cdot y) = x \end{array}$$

TWO-VALUED BOOLEAN ALGEBRA

Table 2.1

Postulates and Theorems of Boolean Algebra

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$
Theorem 2	(a)	$x + 1 = 1$	(b)	$x \cdot 0 = 0$
Theorem 3, involution		$(x')' = x$		
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Theorem 4, associative	(a)	$x + (y + z) = (x + y) + z$	(b)	$x(yz) = (xy)z$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a)	$(x + y)' = x'y'$	(b)	$(xy)' = x' + y'$
Theorem 6, absorption	(a)	$x + xy = x$	(b)	$x(x + y) = x$

OPERATOR PRECEDENCE

- Parentheses

$$(\dots) \bullet (\dots)$$

- NOT

$$x' + y$$

- AND

$$x + x \bullet y$$

- OR

$$x [y + z \overline{(w + x)}]$$

$$\overline{(w + x)}$$

$$\overline{(w + x)}$$

$$z \overline{(w + x)}$$

$$y + z \overline{(w + x)}$$

$$x [y + z \overline{(w + x)}]$$

BOOLEAN FUNCTIONS

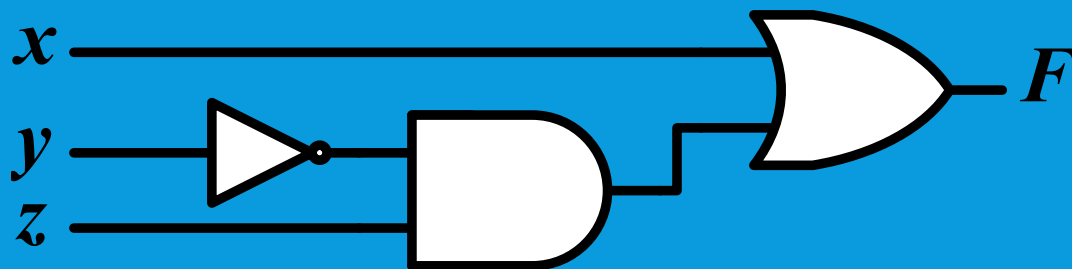
- Boolean Expression

Example: $F = x + y'z$

- Truth Table

All possible combinations
of input variables

- Logic Circuit



x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1